

02LQD Specification and Simulation of Digital Systems

2020/02/14

1. Combinational circuit design (3 points)

A combinational device has a 4-bit input signal x , a 1-bit input signal sel , a 3-bit output y and a 1-bit output err . x and y are unsigned binary numbers. The device computes the discrete base-2 logarithm of x and outputs it on y . When $sel=0$, $y = \lceil \log_2 x \rceil$, when $sel=1$, $y = \lfloor \log_2 x \rfloor$. Whenever y is not a valid result, $err = 1$.

Reminder: $\lceil R \rceil$ is the smallest integer S such that $S \geq R$. Examples: $\lceil 3.1 \rceil = 4$, $\lceil 3.0 \rceil = 3$.

Dually $\lfloor R \rfloor$ is the largest integer S such that $S \leq R$. Examples: $\lfloor 3.9 \rfloor = 3$, $\lfloor 3.0 \rfloor = 3$.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

A Moore FSM has a 1-bit input X and a 1-bit output Z . The FSM removes the first 1 from every sequence of 1s on its input and replaces it with a 0.

Example:

$X = 0011101000111110$

$Z = 0001100000011110$

1. Draw the state diagram (4 points)
2. Design the FSM resorting to the VHDL behavioural description style with 2 processes (4 points).
3. Design the FSM resorting in Verilog with 2 always blocks (4 points).

Reset is asynchronous.

3. FSM-D design (15 points)

Design a FSM-D for a simple data encryption/decryption device. The device has 3 1-bit inputs b , e and d and a 16-bit signed input I . When $b=1$, no matter the values of e and d , the device stores the data on I . This data represents an offset value. When $e=1$ and $b=0$, no matter the value of d , the device encrypts the data it receives on input I adding to it the offset and outputs the result on a 16-bit signed output O . When $d=1$ and both $b=0$ and $e=0$, the device decrypts the data it receives on input I subtracting from it the offset and outputs the result on a 16-bit signed output O . Reset is asynchronous.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (7 points)
2. Design a FSM-D (8 points):
 - create 2 processes for behavioural datapath description from the HLSM state diagram
 - create 2 processes for behavioural controller description from the HLSM state diagram

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

no later than Monday February 17 11.59 pm

- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.